Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

863
3 0.35%
Jan 13th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

975
3.00 0.31%
Jan 13th, 2023

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

745
3 0.40%
Jan 13th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

118.90
0.00 0.00%
Jan 13th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

132.45
0.00 0.00%
Jan 13th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

934
-22 -2.30%
Week Ending Jan 13th, 2023

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2305 (Jan) RMB/t (3pm close)

881.00
25.50 2.98%
Jan 13th, 2023 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
February 23 USD/dmt

125.85
3.50 2.86%
Jan 13th, 2023 (5.30 pm Print)

SHFE Rebar
RB2305 (Jan) RMB/t

4173
37 0.89%
Jan 13th, 2023 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

17.79
-0.24 -1.33%
Jan 12th, 2023

C5, W. Australia - Qingdao USD/t

7.10
0.11 1.57%
Jan 12th, 2023

Steel Price

Steel Rebar (China Domestic) RMB/t

4140
20 0.49%
Week Ending Jan 13th, 2023

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

130.19
2.32 1.81%
Week Ending Jan 13th, 2023

Steel Inventory in China
million tonnes

11.07
1.08 10.83%
Week Ending Jan 13th, 2023

Steel Price

Steel HRC (China Domestic) RMB/t

4230
60 1.44%
Week Ending Jan 13th, 2023

IRON ORE PORT STOCK INDEX (IOPI)

Jan 13th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	863	3	0.3%	771	862	643	1083	119.36	1.11	0.9%	102.47	119.94	82.29	159.06
IOPI58	58% Fe Fines	745	3	0.4%	687	732	604	907	103.62	1.03	1.0%	91.83	102.13	77.29	134.15
IOPI65	65% Fe Fines	975	3	0.3%	898	1015	811	1265	135.37	1.20	0.9%	119.91	142.02	108.78	187.25

IRON ORE SEABORNE INDEX (IOSI)

Jan 13th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	118.90	0.00	0.00%	103.25	122.05	83.90	159.45
IOSI65	65% Fe Fines	132.45	0.00	0.00%	113.53	139.84	94.45	185.82

MARKET COMMENTARY

DCE iron ore futures rose by 3.4 % today, the main contract closed 881.The traders' willingness to ship is general.The steel mills are wait and see .The overall trading sentiment of the market is worse.PBF at Shandong port deal 860-872 yuan/mt,increase 10-29 yuan/mt; PBF at Tangshan port dealt 872-875 yuan/mt,increase 11-14 yuan/mt. As of January 13, the inventory of 35 ports tracked by SMM totaled 130.19 million tons, up from 2320000 tons last week and down from 198.9 million tons last year. The average daily port dredging volume of imported ore in this period increased by 133000 tons to 3.144 million tons on a weekly basis. The price of imported ore was relatively strong this week, but speculative demand picked up and some market participants were still active. From the perspective of supply side, the port centralized unloading before the Spring Festival, and the arrival volume of China's main ports this week increased by 6.1881 million tons on a month-on-month basis, with an obvious increase. On the demand side, the average output of molten iron rebounded this Sunday, driving the volume of port dredging. In the future, the replenishment of steel mills is nearing the end, traders' willingness to trade may weaken, and the

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Jan 13th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	934	-22	-2.3%	884	1039	770	1405	123.57	-0.21	-0.17%	113.19	139.58	94.72	199.13

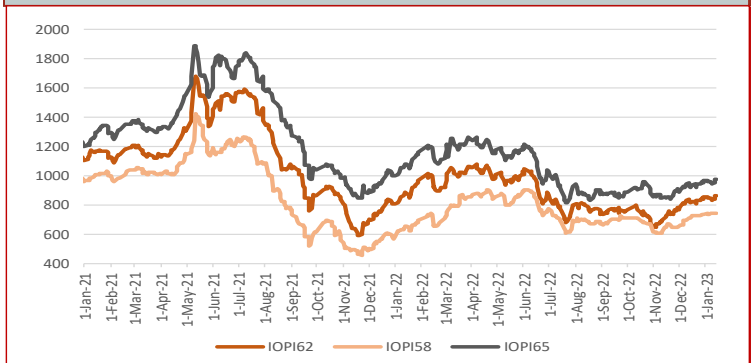
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Jan 13th, 2023				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1039	0.9%	779	1645	153.42	2.94%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	1085	1.9%	780	1630	160.22	3.97%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	760	1.3%	620	1310	112.23	3.41%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1062	0.9%	800	1752	156.82	2.93%	117.19	272.32

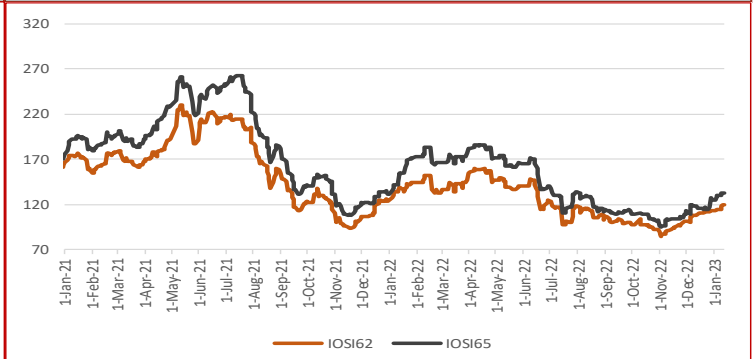
Week Ending Jan 13th, 2023	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	932.40	1.58%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.77208

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Jan 13th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	September	October	November	December	MTD	QTD	YTD	September	October	November	December	MTD	QTD	YTD
IOPI62	62% Fe Fines	759	736	714	822	771	771	862	101.14	95.56	92.18	109.43	102.47	103.03	119.94
IOPI58	58% Fe Fines	697	683	635	711	687	686	732	93.56	89.34	82.39	95.17	91.83	92.23	102.13
IOPI65	65% Fe Fines	875	918	865	935	898	901	1015	117.19	120.08	112.37	124.90	119.91	120.98	142.02

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Jan 13th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	September	October	November	December	MTD	QTD	YTD
IOSI62	62% Fe Fines	101.12	95.80	93.05	108.92	103.25	104.27	122.05
IOSI65	65% Fe Fines	111.57	106.70	102.40	116.83	113.53	115.19	139.84

FREIGHT RATES

Jan 12th, 2023		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	7.10	0.11	1.57%	3.57	16.77
Tubarao - Qingdao	C3	17.79	-0.24	-1.33%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Jan 13th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	September	October	November	December	MTD	QTD	YTD	October	October	November	December	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	873	877	811	922	884	888	1039	112.75	109.81	100.83	117.06	113.19	114.47	139.58

IRON ORE INDEX PREMIUMS/DISCOUNTS

Jan 13th, 2023		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-118	-13.67%
IOPI65	65% Fe Fines	112	12.98%

Jan 13th, 2023		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	13.55	11.40%

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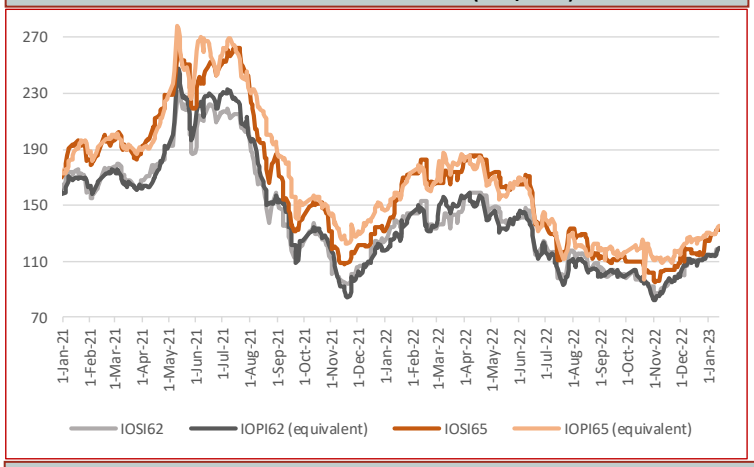
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Jan 13th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	806	3	-57
SIMEC Fines	735	3	-128
PB Fines	831	3	-32
Newman Fines	832	3	-31
MAC Fines	813	3	-50
Jimblebar Blended Fines	728	3	-135
Carajas Fines	1043	3	180
Brazilian SSF	827	3	-36
Brazilian Blend Fines	845	3	-18
RTX Fines	746	3	-117
West Pilbara Fines	774	3	-89

Jan 13th, 2023	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	114.40	0.00	-4.50
SIMEC Fines	110.90	0.00	-8.00
PB Fines	115.15	0.00	-3.75
Newman Fines	118.05	0.00	-0.85
MAC Fines	115.15	0.00	-3.75
Jimblebar Blended Fines	107.55	0.00	-11.35
Carajas Fines	148.45	0.00	29.55
Brazilian SSF	122.65	0.00	3.75
Brazilian Blend Fines	124.30	0.00	5.40
RTX Fines	108.80	0.00	-10.10
West Pilbara Fines	113.15	0.00	-5.75

Jan 13th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	745	3	0
FMG Blended Fines	754	3	9
Robe River	755	3	10
Western Fines	758	3	13
Atlas Fines	752	3	7
Yandi	738	3	-7

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	17.00	-2.00
	High Grade Fe 63 - 64%	31.00	0.00
	High Grade Fe 64 - 65%	31.00	0.00
	High Grade Fe 65 - 65.5%	31.00	0.00
	Low Grade Fe	31.00	1.00
1% Alumina	High Fe Grade Al <2.25%	122.00	3.00
	High Fe Grade Al 2.25-4%	24.00	-5.00
	Low Fe Grade Al <2.25%	8.00	2.00
	Low Fe Grade Al 2.25-4%	46.00	5.00
1% Silica	High Fe Grade Si <4%	38.00	0.00
	High Fe Grade Si 4-6.5%	34.00	2.00
	Low Fe Grade	35.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	13.00	-1.00
	High Fe Grade 0.115%<P<0.15%	13.00	-1.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	0.75	0.00
	High Grade Fe 63 - 64%	1.00	-0.25
	High Grade Fe 64 - 65%	1.00	-0.25
	High Grade Fe 65 - 65.5%	1.00	-0.25
1% Alumina	High Fe Grade Al <2.25%	1.75	-0.50
	High Fe Grade Al 2.25-4%	0.25	0.00
1% Silica	High Fe Grade Si <4%	0.75	0.00
	High Fe Grade Si 4 - 6.5%	0.50	0.25
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	1.50	0.25
	High Fe Grade 0.115%<P<0.15%	5.25	0.00

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-5.00	0.00	Jingtang	-20.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-10.00	0.00

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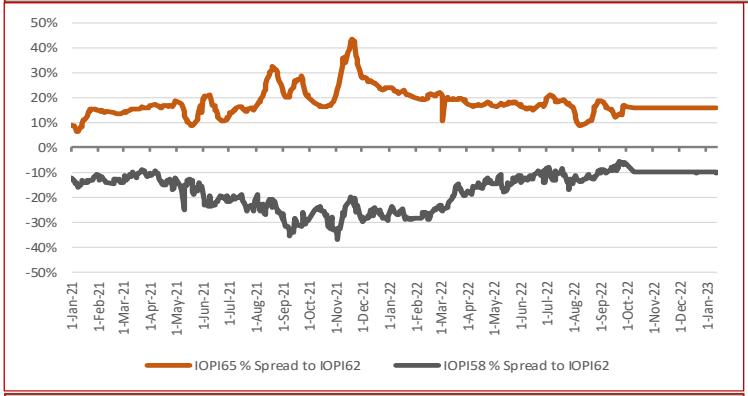
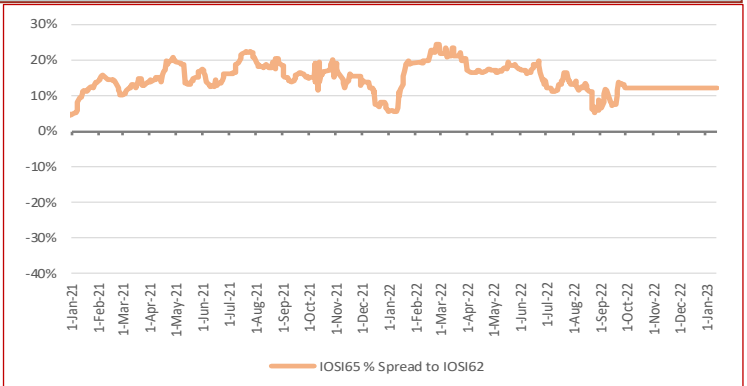
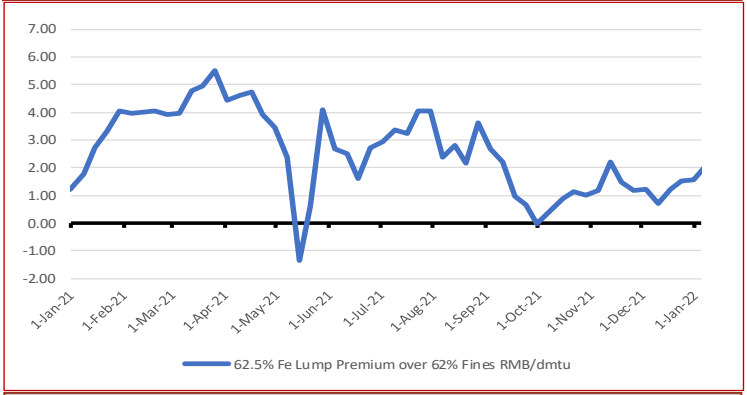
IRON ORE INDEX PREMIUMS/DISCOUNTS

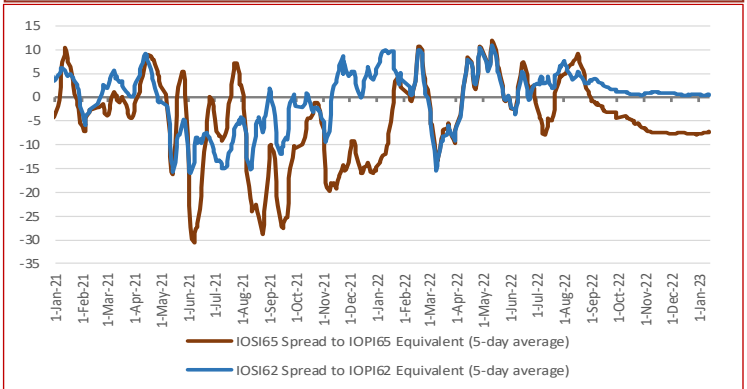




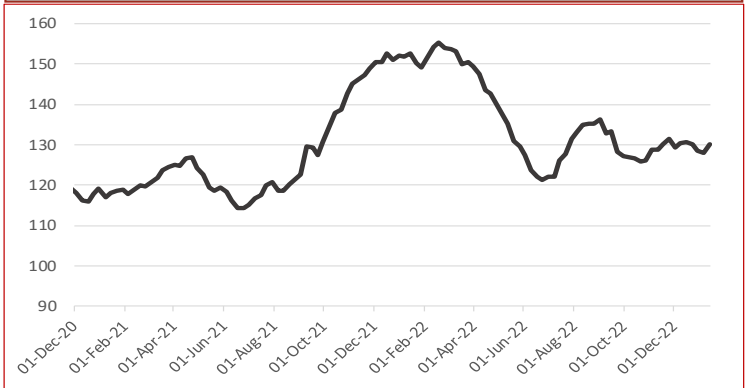
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



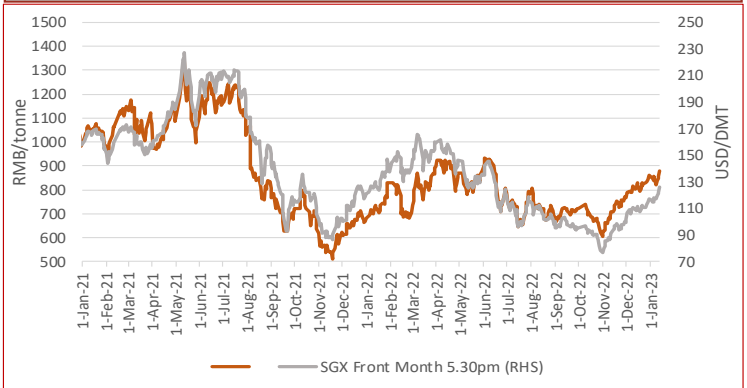
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

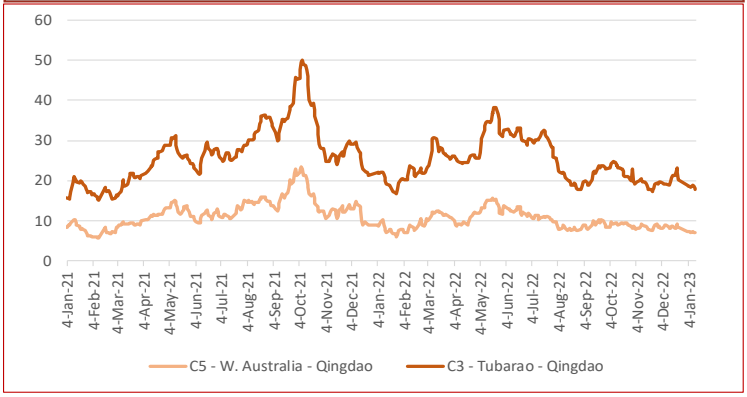
Week Ending Jan 13th, 2023

Province	This week	Change %	Low [2]	High [2]
Jingtang	11.91	-2.93%	9.84	19.20
Qingdao	25.05	-1.92%	9.41	26.24
Caofeidian	11.67	5.14%	9.05	16.29
Tianjin	10.73	2.78%	7.14	12.97
Rizhao	17.83	4.03%	9.44	19.26
Total (35 Ports)	130.19	1.81%	98.80	155.39

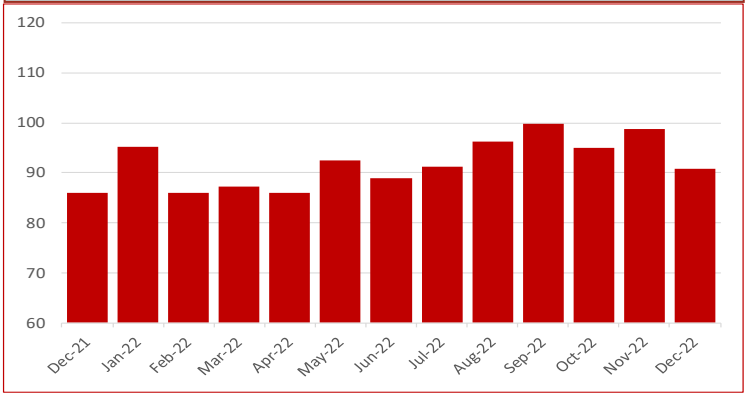
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Jan 13th, 3pm close			Jan 13th, 5:30pm		
Contract	I2305	Change	Change %	Feb. 23	Change	Change %
Closing Price	881.00	25.50	2.98%	125.85	3.50	2.86%
Vol traded ('000 lots)	74.85	10.88	17.00%	6.82	3.12	84.09%
Open positions ('000 lots)	95.33	2.35	2.52%	27.71	0.34	1.26%
Day Low	854.0	10.50	1.24%	121.90	1.40	1.16%
Day High	883.0	21.50	2.50%	126.40	3.15	2.56%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2023/1/13	Change	Change %
ReBar HRB400 ϕ18mm	4,140	20	0.49%
Wirerod Q300 ϕ6.5mm	4,400	20	0.46%
HRC Q235/SS400 5.5mm*1500*C	4,230	60	1.44%
CRC SPCC/ST12 1.0mm*1250*2500	4,570	20	0.44%
Medium & Heavy Plate Q235B 20mm	4,210	60	1.45%
GI ST02Z 1.0mm*1000*C	4,780	0	0.00%
Colour Coated Plate	7,500	-30	-0.40%

CHINESE STEEL MILL PROFITABILITY

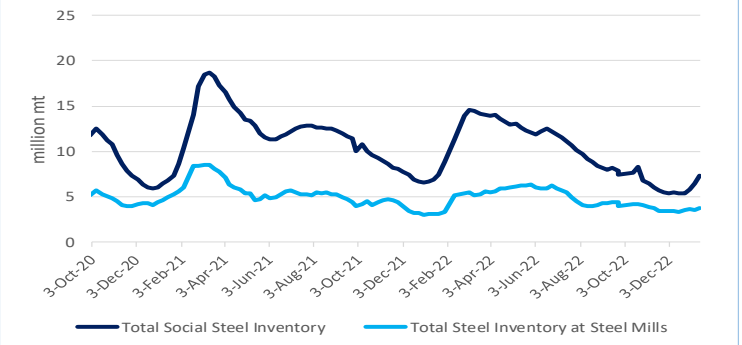
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	118.90	5	Mmi CFR Equivalent index for 1st Feb
Coke	2,960	-110	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	3,120	0	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,756	9	Q234, incl. tax
Rebar cost - Blast furnace	4,278	10	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-78	100	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,233	7	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	-13	43	based on Shanghai prices, incl. tax

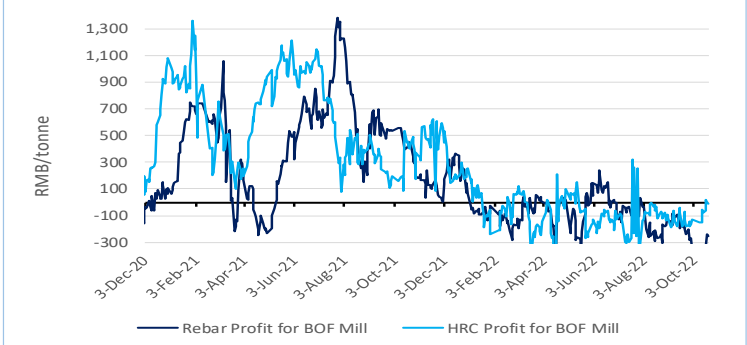
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

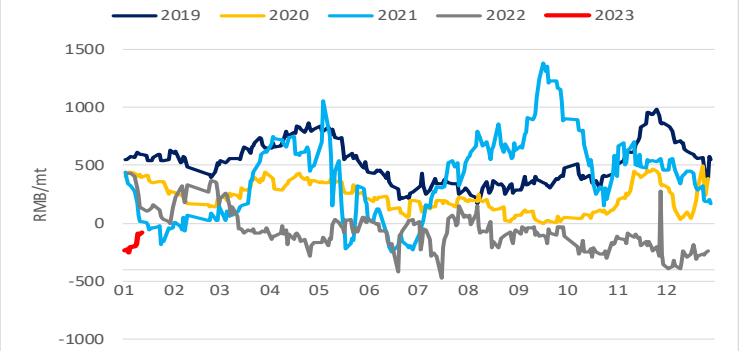
CHINESE STEEL INVENTORIES



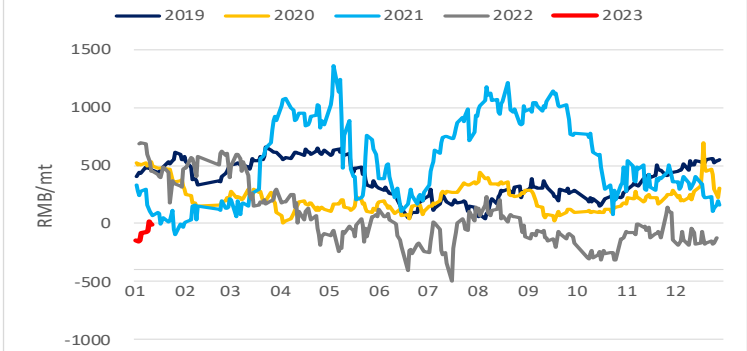
CHINESE STEEL MILL PROFITABILITY



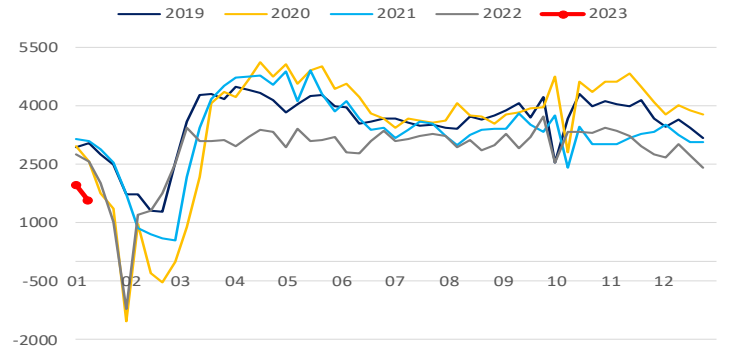
CHINESE STEEL MILL PROFITABILITY—Rebar



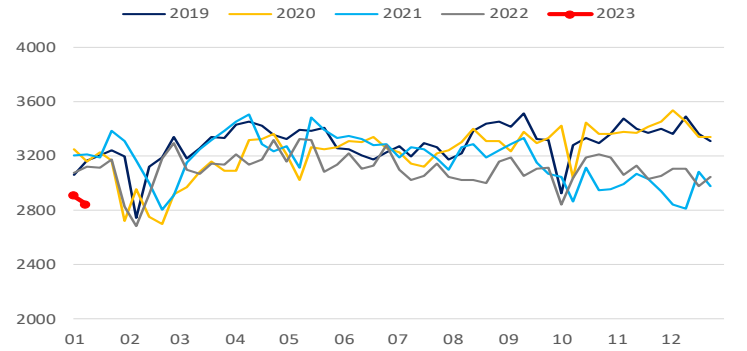
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



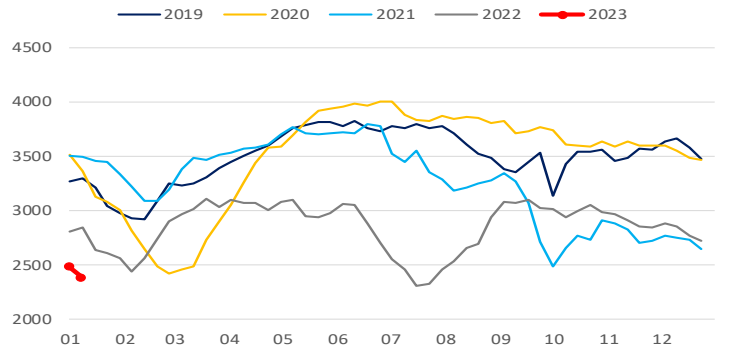
CHINESE STEEL CONSUMPTION—Rebar



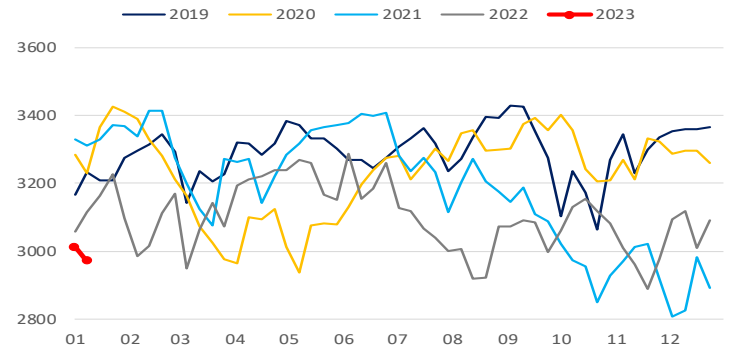
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Jan 13th, 2023	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Jan 13th, 2023	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

CONTACT US

MMI Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7629 E: prices@mmiprices.com
SMM Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7630 E: service.en@smm.cn
SMM Shanghai Office Details: 9th FL, Building 9, Lujiazui Software Park, No.20, Lane 91, Pudong, Shanghai Tel: +86 021 5155 0306 E: service.en@smm.cn

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